



TEXAS A&M UNIVERSITY
ROBOTICS TEAM & LEADERSHIP EXPERIENCE

Week 3: Electromechanical Design

Overview/Outline

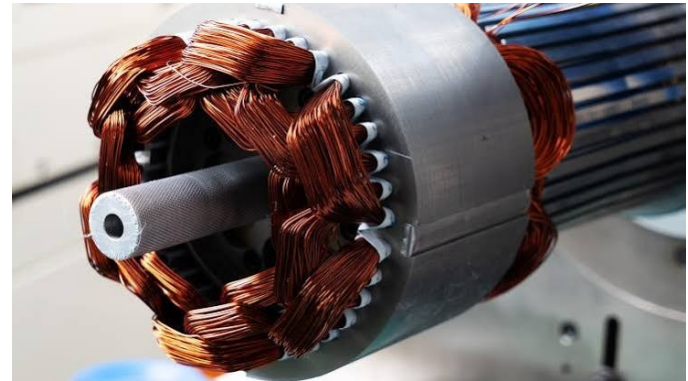


- Part Selection
- Batteries
- Wiring and Connectors
- DC Motors
- Common Electromechanical Devices
- Sensors
- Control and Communications
- Combat Robot Electronics

What is an electromechanical component



- Device that converts electrical energy into mechanical energy or visa versa.
- examples: motors, generators, switches
- They often are part of larger systems and interface with non-electromechanical components





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Part Selection

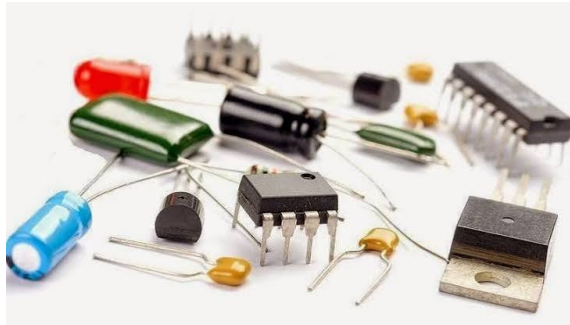
Selecting The Right Components



Every component should satisfy 3 requirements

- Performance - Does it meet the robot's needs?
- Compatibility - Does it work with the rest of the system?
- Constraints - Does it fit within the weight and budget?

Make sure your parts work together, a robot is a complete system. If one component doesn't match, the entire system can fail.



Budget



Every robot has a limited budget, so you must balance performance, reliability, and cost when selecting components.

Ways to Save Money:

- Reuse electronics from previous robots.
- Use standard hardware and connectors.
- Compare prices from multiple suppliers.
- Design around readily available components.
- Buy spare parts for items that commonly break



Every engineering project has constraints, and budget is one of the biggest. Spending twice as much doesn't necessarily make your robot twice as good. The goal is to get the best value, not simply the cheapest or most expensive parts.

Sourcing

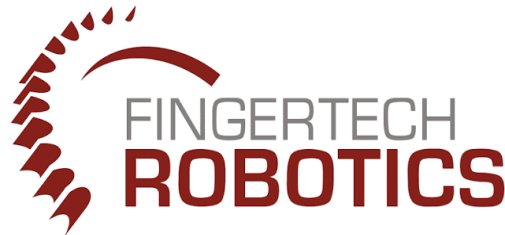


When creating a robot, it's important to purchase components from reliable suppliers. Choosing trusted vendors increases the likelihood of receiving quality parts.

Examples:

- FingerTech Robotics
- Palm Beach Bots
- Amazon
- McMaster-Carr

McMASTER-CARR®



Pugh Table



A Pugh Table is a tool used to compare multiple design options against a set of evaluation criteria. Helping engineers make objective, data driven decisions.

Each option is rated with a +, 0, or – relative to a baseline design. After applying the criterion weights, the option with the highest overall score is selected.

		Option 1	Option 2	Option 3	Option 4
Criteria	Weight	Home	Frank's	Delite	Mongoose
Price	2	0	+	-	--
Diversity	1	0	-	+	++
Food quality	1	0	+	-	+
Distance	2	0	++	+	-
Allergens	2	0	-	++	+
Sally's favorite	1	0	+	0	0
Ben's favorite	1	0	0	+	0
Mike's favorite	1	0	0	+	0
Jen's favorite	3	0	0	0	+
+		0	8	9	8
0		13	5	4	3
-		0	3	3	6
Total		0	5	6	2



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Batteries

(not an electromechanical device, but almost always
needed with one)

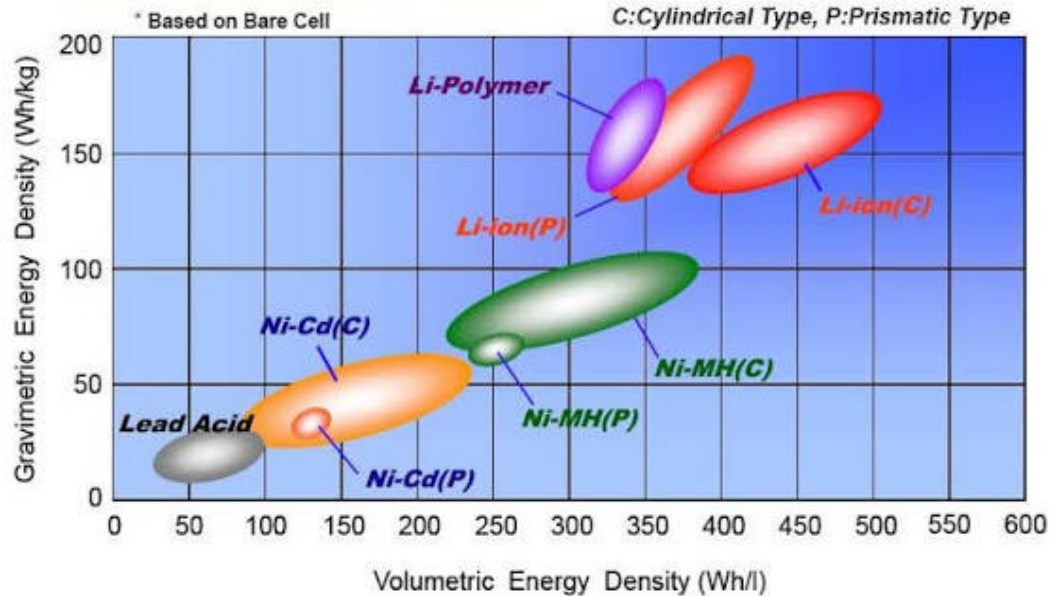
Common Battery Chemistries



- Lead-Acid
 - Cheap, High peak current, Heavy, Large form factor
- Lithium Ion (Li-ion)
 - Highest energy density, Long cycle life
- Nickel Metal Hydride (NiMH)
 - Robust, Compared to lithium: Cheaper, lower energy density
- Lithium Iron Phosphate (LiFePO₄)
 - Very safe, Long cycle life, Thermally stable, lower energy density



Energy Density Chart



Sadly LiFePO₄ not included in this graphic :(

LiPo Basics

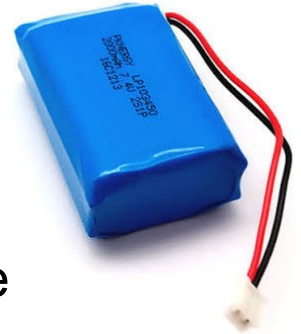


What is a LiPo Battery?

- Lithium Polymer (LiPo) battery is a lightweight high energy battery famous for providing in small sizes
- Has a high discharge current and is a standard battery for RC toys, drones, and mobile devices.

Advantages:

- High power output
- Lightweight
- recharge quickly
- Many sizes available



Disadvantages:

- Overcharging can damage the battery
- Fire risk if punctured
- Require proper chargers and storage

Understanding Battery Specifications



Example: 1S, 3.7V, 150mAh, 25C

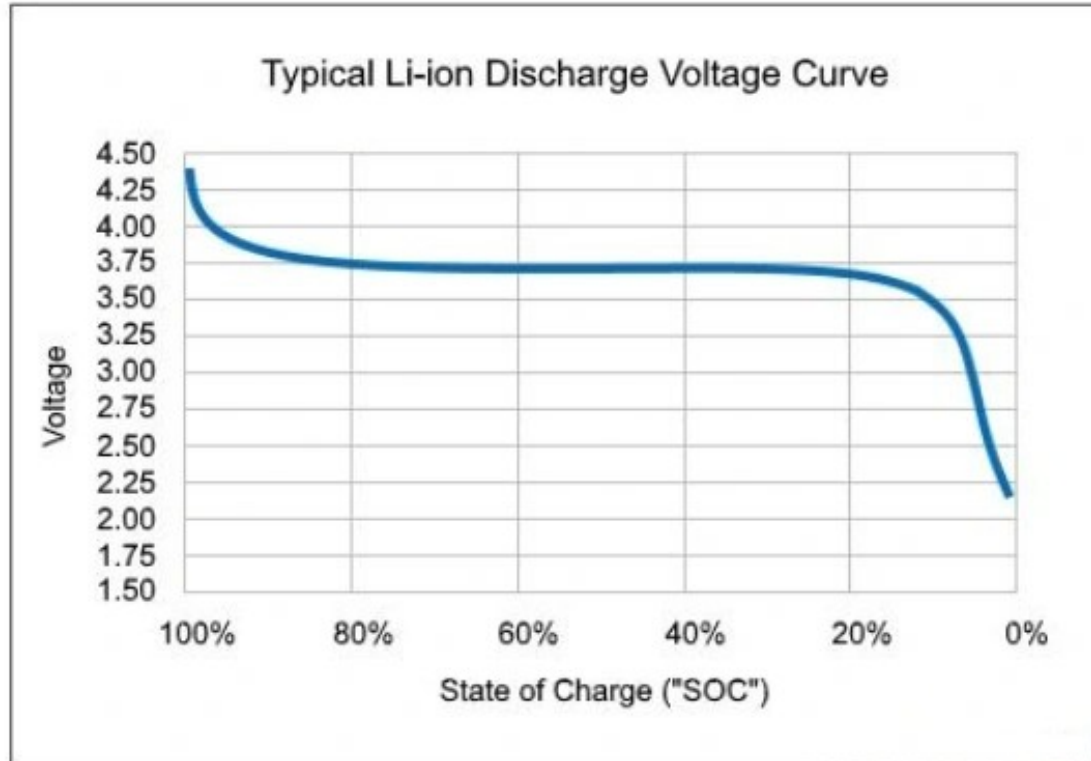
- Cell Count (S): Each LiPo cell comes with a nominal 3.7V, max 4.2V (starts), and min **3.0V (anything below is unsafe)**.

1S (1 Cell) = 3.7V, 2S = 7.4V, 3S = 11.1V,
4S = 14.8V

- Capacity (mAh): 150 mAh means how much energy is stored. Higher capacity = longer runtime + heavier + larger
- C-Rating: Specifies the maximum safe constant current draw
Maximum current = Capacity x C-Rating, 150 mAh = .15 Ah
Ex. Max Current = .15 x 25 = 3.75 A

Capacity adds in parallel, Voltage adds in series

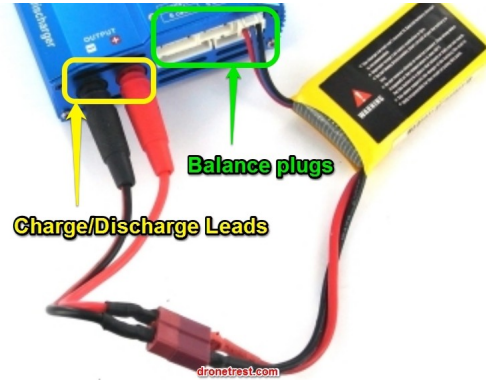
Discharge Curve



LiPo Charging



- These batteries are made from multiple cells
- Each cell needs to be kept at the same voltage while charging
- 2 plugs: balance leads and charge leads



- Select the correct battery type and cell count
- Select the current you charge at
- Should charge at 1C!
- Bring down to storage voltage when not using
- Never charge unattended

LiPo Safety



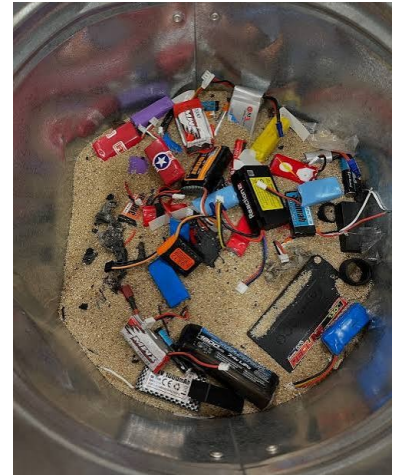
- Always use a LiPo bag when charging!!
- Don't store batteries when they aren't at storage voltage.
- If voltage drops below 3V on a cell, don't use it again
- Always inspect batteries after use.
- If punctured or puffy, let one of us know to dispose it



LiPo Fire



- Usually caused by damage, over charging, or overheating.
 - If you see or smell smoke, notify the officer in the lab immediately
1. Power off charger and disconnect battery
 2. Place battery in sand bucket
 3. Smother with other sand bucket
 4. Take outside





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Wiring and Connectors

Solid Core vs. Stranded Core

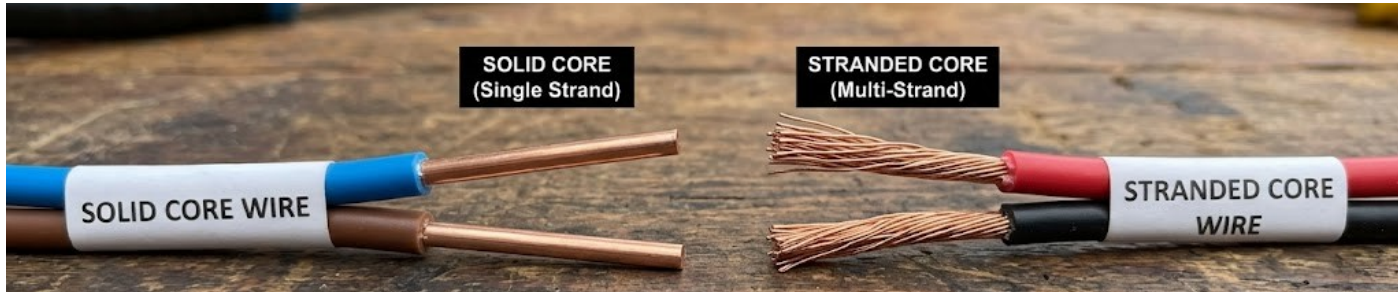


Solid:

- Cheaper
- Higher Current Capacity
- Effectively Male Pins

Stranded:

- More resistant to vibrations
- More flexible



Wire Gauge



Lower # = Thicker Wire, Higher Current, Lower Resistance

Ranging from 40 AWG to 0000 AWG
(Strongest)

What we have and example of power:

- 12 AWG (20 Amps), Ex. Outlets
- 18 AWG (10 Amps), Ex. Lamps
- 22 AWG (3-5 Amps), Ex. Sensor signals



Chart for Reference: [https://www.engineeringtoolbox.com/wire-gauges-d_419.](https://www.engineeringtoolbox.com/wire-gauges-d_419.html)

[html](https://www.engineeringtoolbox.com/wire-gauges-d_419.html)

Current Draw



Current is the amount of electrical charge flowing through a circuit. If Voltage = water pressure, then Current = amount of water flowing through the pipe.

Every component draws current, for example:

- Drive motors = 6A each
- Weapon motor = 15A
- Servo = 1A
- Receiver = 0.2A

$6 + 6 + 15 + 1 + 0.2 = 28.2\text{A}$, The battery should be able to safely provide this much current and always leave extra margin

Common Connectors



- XT Series (XT30, XT60, XT90)

- High-current electrical plugs used for RC models
 - XT# represents how many amps they can handle
 - XT30 (30A peak) is best for 1-3 lb bots



- Bullet Connectors (2mm-8mm)

- Cylindrical electrical terminals used to connect and disconnect single wires easily
 - Robotics/RC: Linking motors, electronic speed controllers (ESCs), and battery packs in drones and robots.



- Servo Connectors (PWM Connectors)

- Standard 3-pin connectors used to link the receiver to speed controllers, microcontrollers, and servos



Common Connectors

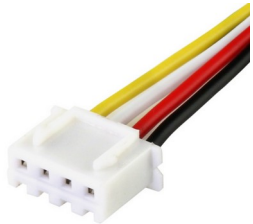


- JST (RCY, XH, PH, SH)

- Electrical connector used in consumer electronics, 3D printers, battery packs, and RC toys



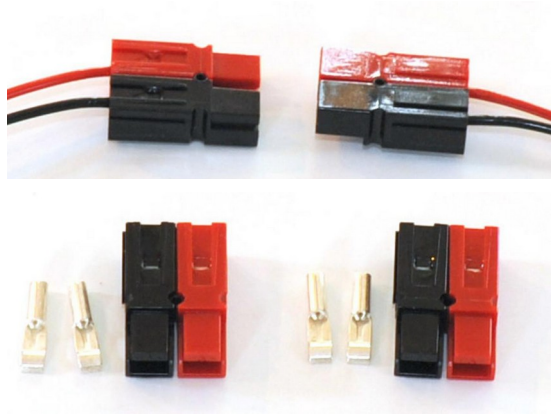
- JST-RCY: 2.5 mm pin spacing, most common type and used for small battery power connections
- JST-XH: 2.5 mm pin spacing, common for LiPo balance leads and breadboards
- JST-PH: 2.0 mm pin spacing, often used for smaller battery connections
- JST-SH: 1.0 mm pin spacing, extremely small and often used for flight controllers and tiny sensors



Other Common Connectors



Ring/Eyelet
Terminal



Anderson
Powerpole



MR30/MR60

Wire Management



Good Wiring ☐



Bad Wiring ☐

Wire Management



Why is it important to have good wiring?

- Bad Spaghetti Wiring = Messy, Uncoordinated, Higher Failure Rate
- Good Wiring = Nice & Neat, Easier Repairs, Lower Failure Rate, Easier Troubleshooting

Ways to improve wiring:

- Crimping, Color Code, Zip ties, Proper Strain Relief





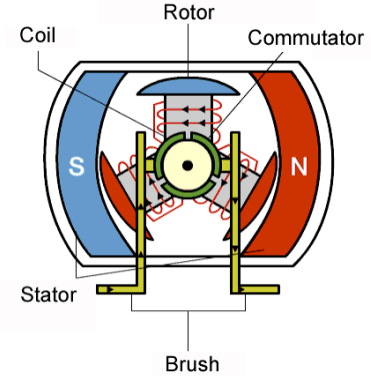
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DC Motors

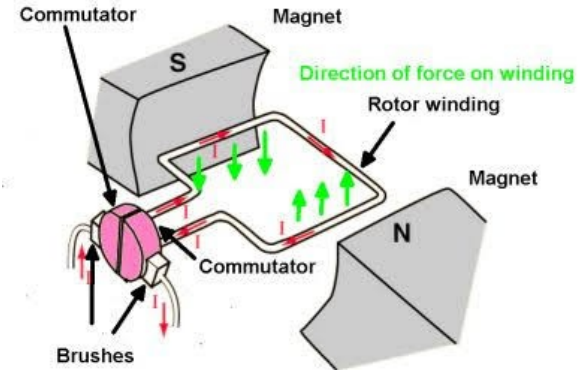
Brushed



- mechanical commutator (brush)
- two wires
- speed proportional to voltage
- cheaper
- Controlled with motor driver/brushed ESC
 - uses PWM to change average voltage



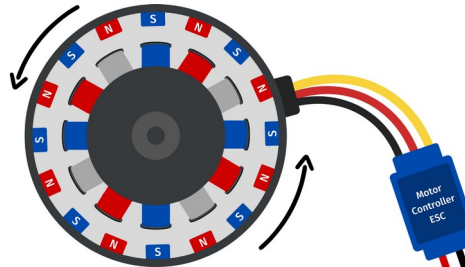
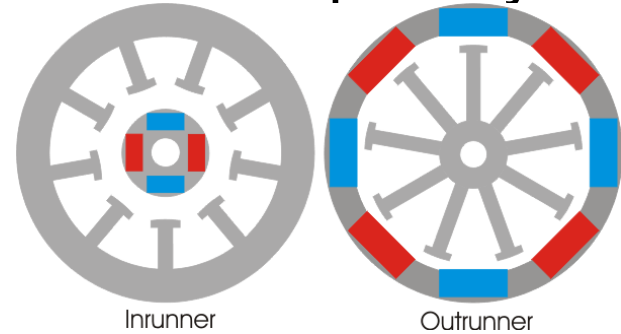
Brush motor principle of operation



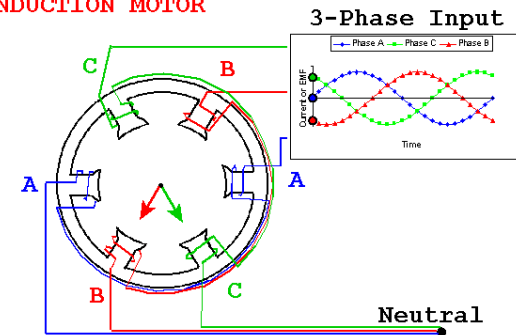
Brushless



- motion is from the timing of the stator coil's polarity
- 3 wires
- inrunner and outrunner types
- needs an ESC to run
- more expensive
- longer life
- higher efficiency



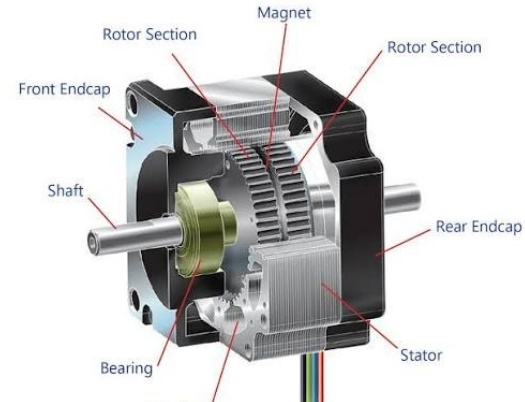
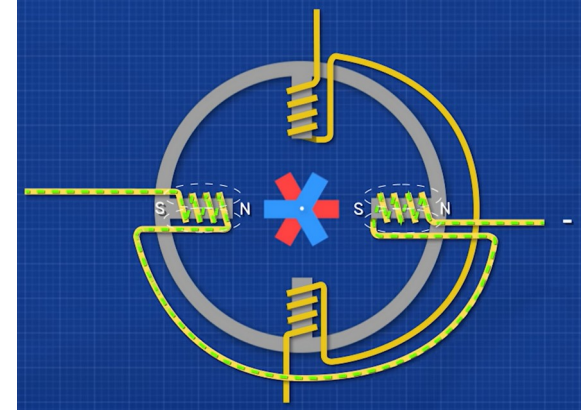
INDUCTION MOTOR



Stepper



- rotates in discrete steps
 - usually 1.8° (200 steps per rotation)
- 4 wires
- requires stepper motor driver
- lower speed, higher torque
- high precision



Linear Motor Model



- Speed is proportional to voltage
 - advertised as the motor's k_v [rpm/V]
 - k_v is the inverse of k_e
- Torque is proportional to current k_T
 - $k_e = k_T$ when in the same units
 - $[V \cdot s / \text{rad}] = [N \cdot m / A]$
- Total voltage includes winding resistance but is negligible at high speeds

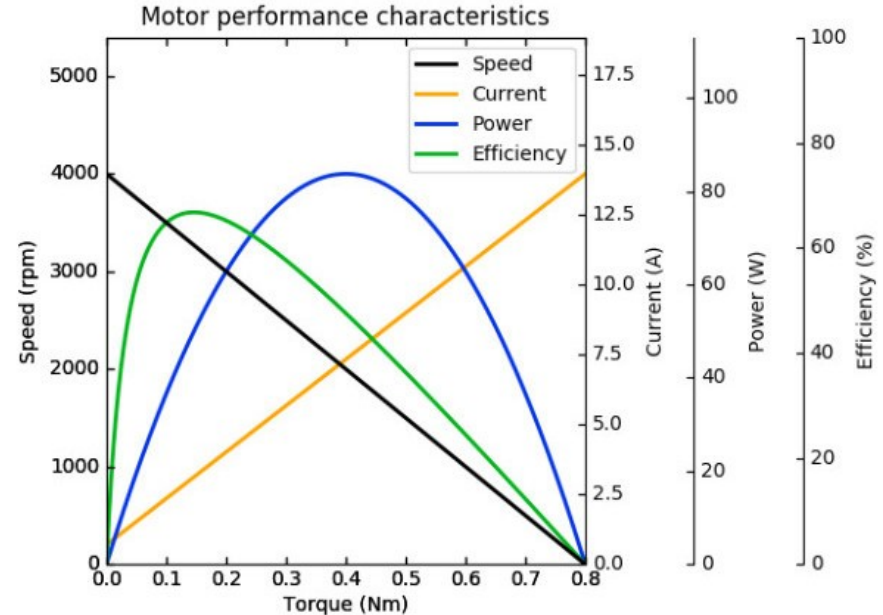
$$V_{\text{applied}} = k_e \omega + I R_w$$

$$T = k_t I$$

Motor Curves



- no-load torque
 - minimum torque required to overcome internal losses
- no-load speed
 - max speed, speed when torque is zero, speed given by KV
- no-load current
 - minimum current
- stall torque
 - max torque
- stall current
 - max current

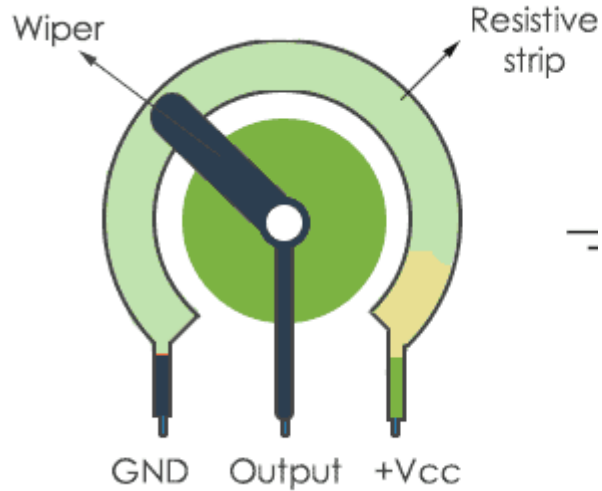




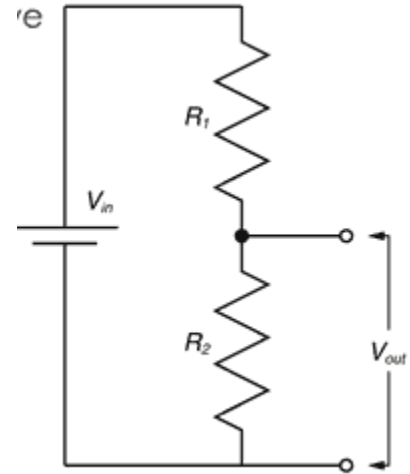
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Other Common Devices

Potentiometer



- Uses a sliding contact to make a variable resistor
- Converts rotation or translation into resistance
- Usually 3 pins



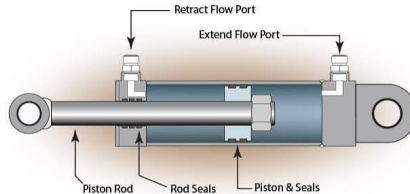
Servo



- Rotates to a specified angle with higher torque
- 3 wires (signal, power, & ground)
- Combination of
 - DC motor
 - potentiometer
 - gearbox
 - control circuit
- Typically limited to 180°
- PWM control (more later)
 - The pulse width encodes the servo's position



Linear Actuators

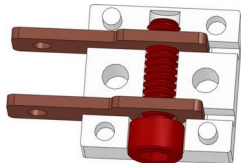
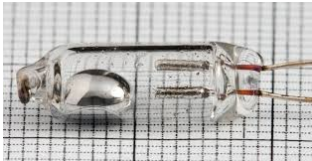
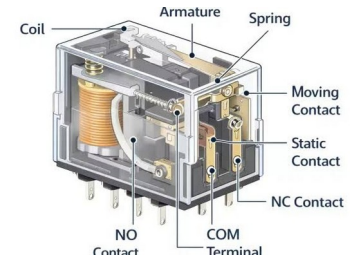
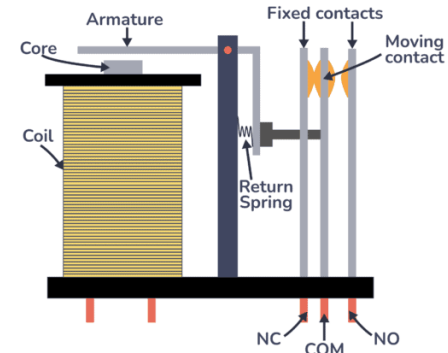
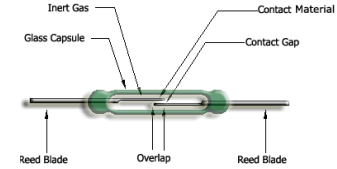


- Creates linear motion
- Usually converts circular motion to linear
 - rack and pinion, belt, and screw drives
- pneumatic and hydraulic options

Switches and Relays



- open and closed circuits
- usually a physical input (mechanical/limit switches)
- Can use magnets or inclination to actuate (reed/mercury switches)
- Relays are a special kind of switch
 - allow you to open or break a high voltage circuit from a distance
 - electromagnet operates switch from a distance





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Sensors

What is a Sensor?



A sensor is a device that measures something in the real world and converts it into an electrical signal that a controller can understand.

Examples of things sensors measure:

- Position
- Distance
- Speed
- Rotation
- Temperature
- Light

Common Sensors



- **Hall Effect**
 - Transduce magnetic fields to electrical signals
- **Accelerometer**
 - Measures acceleration
- **Gyroscope**
 - Measures the change in rotation angle per unit of time
- **IMU (Inertial Measurement Units)**
 - Accelerometer + Gyroscope, Measures acceleration and rotation
- **Rotary Encoder**
 - Measures rotation of of a shaft
- **Limit Switch**
 - Detects position by physical contact
- **Ultrasonic Distance**
 - Measures distance through ultrasonic waves
- **Torque Sensor**
 - Measures torque on a rotating system



Precision vs. Sensitivity



Precision - How close repeated measurements are to each other (Repeatability)

Ex. All darts landing together



Sensitivity - Smallest detectable change

Ex. Scale detects by 0.1 g



Sensor Characteristics: Resolution



- Definition
 - The smallest unit your sensor can detect.
 - 0th order uncertainty is $\frac{1}{2}$ your resolution
 - for the scale with 0.01g resolution it has a 0th order uncertainty of $\pm 0.005\text{g}$
- Units
 - same as the measured quantity



Sensor Characteristics: Range/Span



- Definition

- The upper and lower limits of what can be measured
- Operating range are the conditions the sensor can function

- Units

- fixed values (0–30 bar)
- usually a trade off, bigger ranges tend to have smaller resolutions

ECO-2 Digital Pressure Gauge



➔ [Request Quote](#)

➔ [ECO 2 Datasheet](#)

➔ [ECO 2 Manual](#)

➔ [ECO 2 EI Manual](#)

Quick Overview

ECO 2 is a compact and economical digital pressure measuring instrument providing medium accuracy at high resolution and repeatability. The piezoresistive pressure transducer as the heart of the instrument has gone through extensive pressure- and temperature tests.

The pressure is measured twice per second and displayed. The top display indicates the actual pressure, the bottom display shows the Max.- ECO 2 has two operating keys. The left key turns on the instrument and is used to select the functions and the pressure units. The right key executes the selected function or unit. The right key is also used to switch between the Max.- and Min.-value.

ECO 2 has 7/16" UNF male thread and is delivered complete with an O-ring seal swivel fitting which allows ECO 2 to be rotated 360°. Only minimal torque is needed to seal up to 300 bar. Process pressure connection is G 1/4" male.

Specifications

Pressure Ranges	0...31 bar 0...300 bar
Accuracy	± 0.5 %FS Typ.
Storage / Operating Temperature	-20...70 °C / 0...50 °C
Protection	IP64

Sensor Characteristics: Accuracy



- Definition
 - quantifies the difference between the measured value and true value
 - systematic uncertainty inherent to your sensor
- Units
 - fixed value (± 0.001 in)
 - relative to reading ($\pm 1\%$ of measured value)
 - relative to full-scale ($\pm 1\%$ of full scale)

ECO-2 Digital Pressure Gauge



- ➔ [Request Quote](#)
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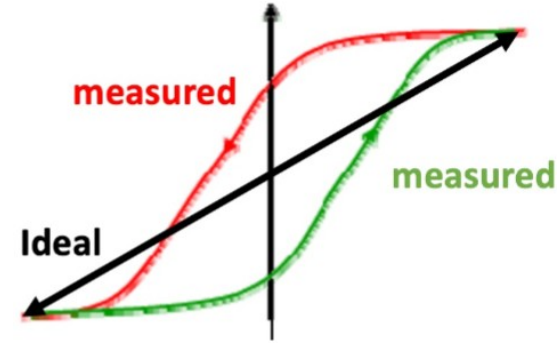
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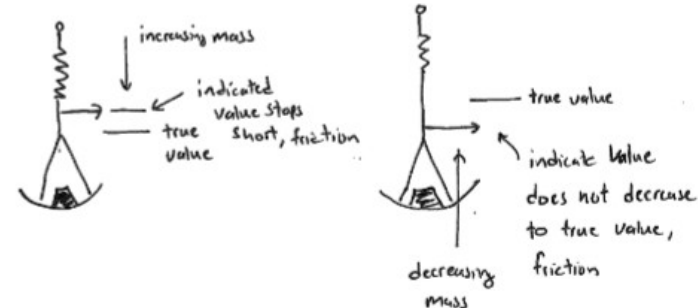
Sensor Characteristics: Hysteresis



- Definition
 - difference in output depending on direction of input
 - get different output for the same input depending on direction
- Units/Expression
 - expressed as a percentage of full-scale



Example: Spring Scale with friction:





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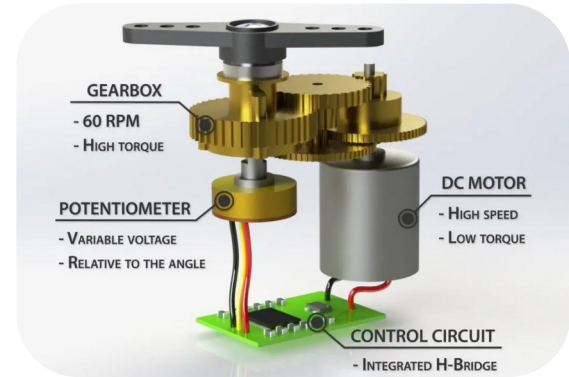
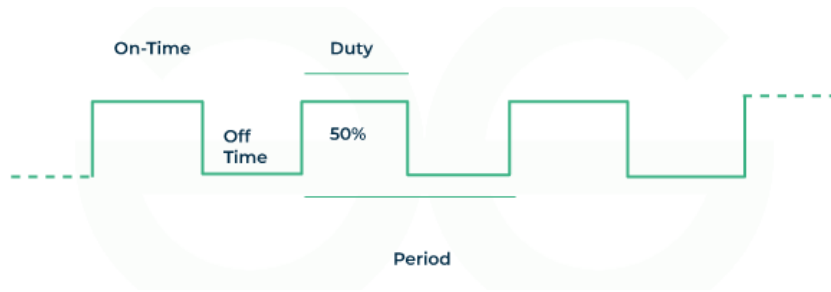
Control and Communications

How do you make anything do what you want?

Closer look at a servo motor



- Has not only power & ground, but a signal wire (PWM)
- encodes the duty cycle % to a position
 - DC%: % of the period that's high



- spins until control circuit reads corresponding potentiometer voltage

Let's generalize



- A **controller** interprets commands and determines what the component should do.
 - Range from onboard computer to a person with an Xbox controller
- Communication signals carry commands between the controller and the component.
- Different components use different communication methods/protocols to exchange information.
 - This determines where the wire(s) must be connected
 - i.g. the PWM wire must go to a PWM supported terminal

Common controllers: Hatchling Review



Besides direct human control, robots commonly use either or both of

Microcontroller

- None or Real Time OS
- Lower power consumption
 - Limited processing
 - Limited memory
- GPIO Peripheral
- Low level languages
 - Single program always running
- Examples:
 - Arduino
 - ESP32/ESP8266
 - STM32



Single-Board Controller

- Full OS (Typically Linux)
- Higher computation ability
- Peripherals
 - GPIO, USB, HDMI, etc
- Higher-level languages
 - Can run multiple programs
- Examples:
 - Raspberry Pi
 - BeagleBoard
 - NVIDIA Jetson Nano



Wired Protocols: Hatchling Review



Communication Protocols			
Name	Pinout	Voltage / Signal	Use Case / Notes
UART	TX, RX (+ GND, Power optional)	3.3V or 5V logic levels	Universal Asynchronous Receiver/Transmitter. Very common serial RX + TX communication. Ports connect in series TX → RX and RX → TX. Simple point-to-point communication.
I2C	SDA, SCL (+ GND, Power)	3.3V or 5V open-drain	Inter-Integrated Circuit. Common for short-distance communication. Two-wire bus allowing multiple devices with unique addresses. Requires pull-up resistors.
CAN Bus	CAN_H, CAN_L (+ GND, Power)	Differential voltage (~5V)	Controller Area Network. Robust vehicle bus standard for microcontroller communication. Voltage difference provides robustness against line loss and interference. Common in automotive and industrial applications.
PWM	Signal, GND (+ Power optional)	3.3V or 5V logic levels	Pulse Width Modulation. Modulation technique encoding messages into pulsing signals. Commonly used for motor control, servos, and ESC communication. Single wire data transmission.

- Different components use different protocols/wirings
- Examples components and protocols (from Hatchling)
 - IMU: UART
 - Accelerometer: I2C
 - PWM: servo & ESC



More information at [TURTLE Knowledge base](#) and [Hatchling week 8 slides](#)



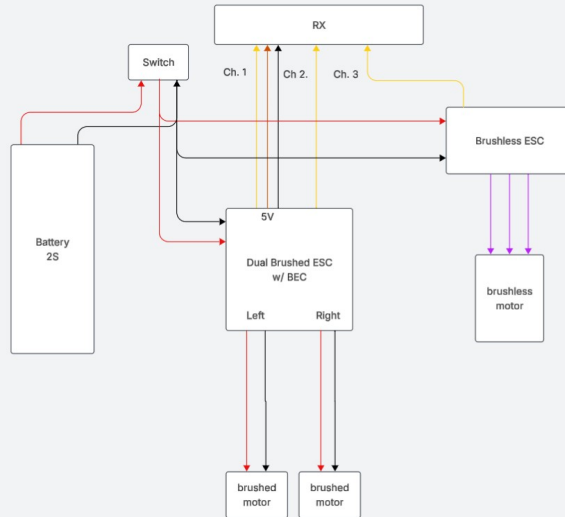
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Combat Robot Electronics

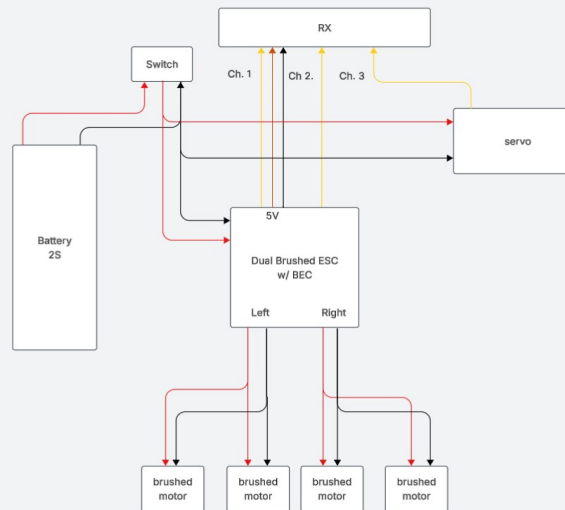
Example Wirings



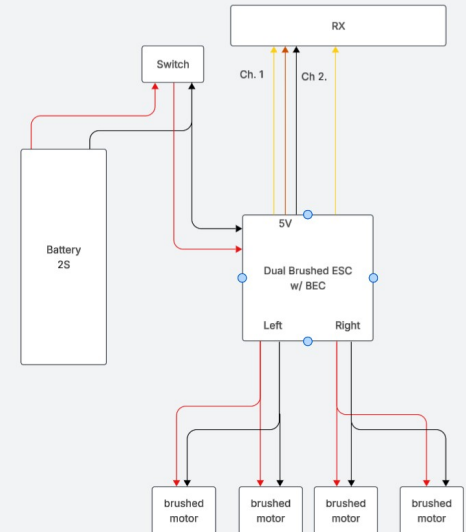
Active Spinning Weapon 2WD



Active Lifting Weapon 4WD



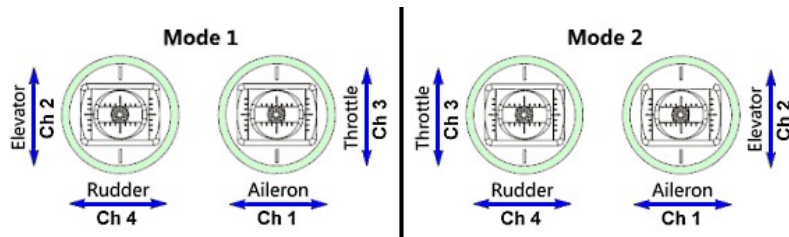
Passive Weapon 4WD



TX/RX FlySky i6



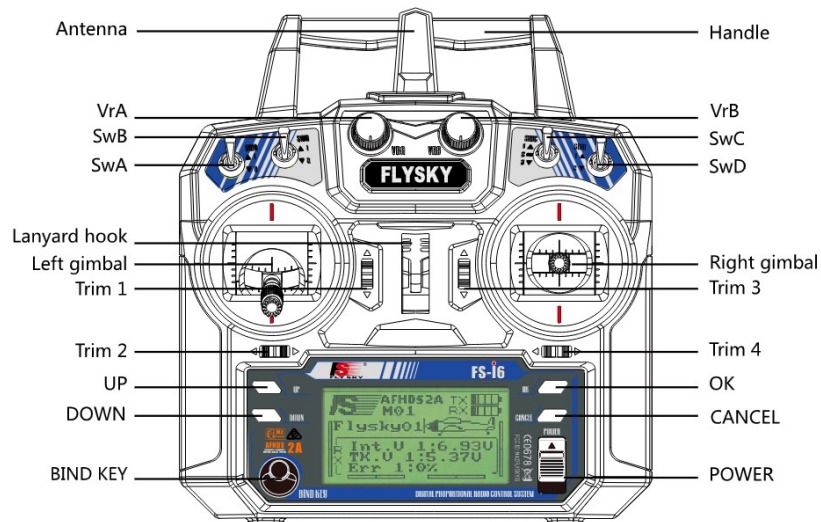
- Each throttle controls a channel (default mode 2)
- The RX sends a PWM signal that varies with throttle position
- Channel mixing: multiple physical inputs control an output.
 - commonly used to make driving easier
- Uses a key to bind TX to RX



More Information



- [manual](#)
- [Combat robot radio guide](#)
(video)

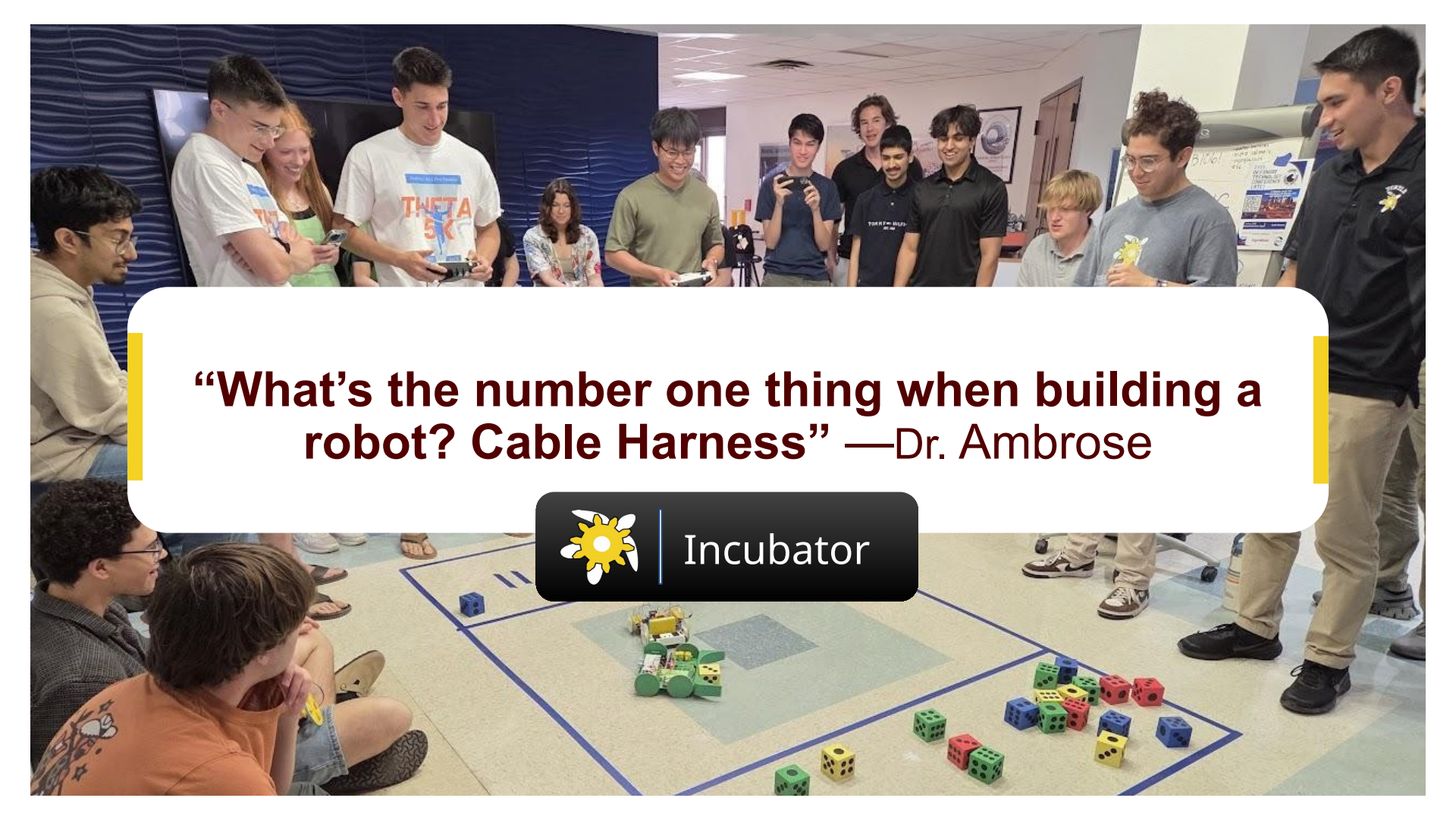




TEXAS A&M UNIVERSITY
ROBOTICS TEAM & LEADERSHIP EXPERIENCE

Next Week

Materials, Manufacturing, and DFM

A group of students are gathered around a small green robot on a floor marked with blue tape. The robot is surrounded by several colorful dice. In the background, a group of students is standing and talking. One student is holding a small yellow object. The scene is set in a classroom or lab with blue acoustic panels on the wall.

“What’s the number one thing when building a robot? Cable Harness” —Dr. Ambrose



Incubator